

Package #141

Implementing a cost-effective battery management system with enhanced monitoring and control capabilities that can detect abnormal charging patterns, prevent overcharging, and isolate cells showing signs of degradation

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'lithium-ion consumer-product battery fire': 5 of 50 CPSC recalls in the retrieved window involve batteries/chargers/power electronics. Recalled products: 'Cooluli Recalls 10-Liter and 15-Liter Minifridges Due to Fire and Burn Hazards'; 'Shengsen Prank Sound Makers Recalled Due to Risk of Serious Injury or Death from Battery Ingestion'; 'A2batt Recalls EEMB Lithium Coin Battery Chargers Due to Risk of Serious Injury or Death from Batter'. LIMITS STAMPED INLINE: a recall is an acknowledged product hazard, NOT an incidence rate; injury lists are associated complaints, not a census (Art. XXI.5 analog).

Mechanism: smart battery management systems that enhance resilience while maintaining energy density

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.